Linear Programming - Tutoring Session Log

October 13, 2026 | Delectation product mix and three-quarter production planning

This is a structured reconstruction of the conversation visible in the current session, preserving the sequence, submitted R code, mathematical constraints, explanations, and results. Assistant responses are condensed for readability; this is not a verbatim export. Individual message timestamps were not available.

Part I - Sandwich product mix

01 USER

Asked how to translate an example make.lp/add.constraint script into a sandwich model, emphasizing learning rather than receiving a finished script. Defined F = Fish-n-Fowl, S = Surf-n-Turf, objective P = 5.50F + 6.95S, and five supply constraints.

```
Tuna: 0.25F + 0.25S <= 200
Beef: 0.4S <= 60
Turkey: 0.5F <= 260
Rye: 2F + S <= 1500
Pumpernickel: F + S <= 800
```

02 ASSISTANT

Explained make.lp(0,2): zero preallocated constraints, two variables. Suggested a consistent order F then S, and invited user to build the first model line.

03 USER

Asked specifically to break down add.constraint().

04 ASSISTANT

Explained add.constraint(model, coefficient_vector, direction, RHS). Tuna translates to c(0.25,0.25), <=, 200. Beef uses c(0,0.4), since F is absent. Emphasized a fixed F,S ordering and practiced the turkey constraint.

05 USER

Submitted the five supply constraints and a first draft of the model.

```
ProductMix.LP <- make.lp(5,2)
add.constraint(ProductMix.LP,c(0.25,0.25),"<=",200)
add.constraint(ProductMix.LP,c(0,0.4),"<=",60)
add.constraint(ProductMix.LP,c(0.5,0),"<=",260)
add.constraint(ProductMix.LP,c(2,1),"<=",1500)
add.constraint(ProductMix.LP,c(1,1),"<=",800)
set.objfn(ProductMix.LP,c(3,2))
lp.control(ProductMix.LP,sense="max")
solve(ProductMix.LP)
```

06 ASSISTANT

Verified all five coefficients and capacities. Corrected make.lp(5,2) to make.lp(0,2), because constraints were being added afterward. Corrected objective to c(5.50,6.95). Explained dimnames(model) <- list(rnames,cnames) and default nonnegativity.

07 USER

Shared the printed LP and solution: profit coefficients 5.5 and 6.95; optimum 520 Fish-n-Fowl and 150 Surf-n-Turf; objective 3902.5. Asked whether this meant $3,902.50 profit.

08 ASSISTANT

Confirmed: 520 Fish-n-Fowl, 150 Surf-n-Turf, and $3,902.50 maximum profit; verified 520 x 5.50 + 150 x 6.95 = 3902.50.

09 USER

Asked how to fill a sensitivity table and interpreted the turkey output: shadow price 11, RHS from 0 to 325.

10 ASSISTANT

Explained binding as using the full resource, and shadow price as marginal change in optimal objective per added unit. Turkey uses 0.5 x 520 = 260 lbs, exactly its limit; therefore binding. Shadow price $11/lb for valid RHS 0-325.

11 USER

Shared sensitivity output for all resources, including scientific-notation ranges.

```
Tuna: shadow price 0; RHS from -1e+30 to 1e+30
Beef: shadow price 17.375; RHS from 0 to 112
Turkey: shadow price 11; RHS from 0 to 325
Rye: shadow price 0; RHS from -1e+30 to 1e+30
Pumpernickel: shadow price 0; RHS from -1e+30 to 1e+30
```

12 ASSISTANT

Explained 1e+30 = 10^30, a solver-scale stand-in for a large bound/infinity. Noted the reported nonbinding lower ranges looked inconsistent with the economic meaning of the model. Listed binding beef/turkey and nonbinding tuna/rye/pumpernickel. Explained that a zero shadow price does not permit reducing inventory below consumption without consequences.

13 USER

Shared sensitivity-analysis R script.

```
plot(ProductMix.LP)
get.sensitivity.obj(ProductMix.LP)
get.sensitivity.rhs(ProductMix.LP)
get.rhs(ProductMix.LP)
get.constraints(ProductMix.LP)
v <- get.sensitivity.rhs(ProductMix.LP)
sprintf("%g",get.sensitivity.rhs(ProductMix.LP)$duals)
rnames <- c("Tuna Used","Roast Beef Used","Turkey Used",
            "Rye Used","Pumpernickle Used",rep("Non-negativity",2))
sprintf("shadow price for %s constraint is %g which is valid for RHS values from %g to %g",
        rnames,get.sensitivity.rhs(ProductMix.LP)$duals,
        get.sensitivity.rhs(ProductMix.LP)$dualsfrom,
        get.sensitivity.rhs(ProductMix.LP)$dualstill)
```

14 ASSISTANT

Explained $duals, $dualsfrom and $dualstill. Stated corrected economically meaningful RHS ranges for the nonbinding resources as tuna 167.5 to infinity, rye 1190 to infinity, and pumpernickel 670 to infinity; recommended investigating why the solver returned very broad ranges. Suggested changing tuna RHS to 100 and resolving to test sensitivity.

15 USER

Asked whether increasing rye slices from 1500 to 1600 provided no financial gain because rye shadow price was zero.

16 ASSISTANT

Confirmed. Rye usage was 1190 of 1500 slices, leaving 310 unused; adding 100 would not improve maximum profit. Suggested mentioning the rye constraint is nonbinding.

17 USER

Asked whether turkey shadow price $11 meant $11 per additional pound.

18 ASSISTANT

Confirmed the marginal $11/lb interpretation while turkey RHS remains within 0-325, other things equal; noted this is marginal value, not purchase price.

Part II - Three-quarter production and inventory LP

19 USER

Introduced Question 2: demands of 3000, 2000, and 4000; initial usable inventory of 1000; regular-time capacity 2700 units per quarter at $40 per unit; unlimited overtime at $60; 20% production defects; 10% spoilage of units left after demand; $15 holding cost for each quarter-ending usable inventory. Variables R1-R3, O1-O3, I1-I3 and P1-P3. Proposed minimizing 40(R1+R2+R3)+60(O1+O2+O3)+15(I1+I2+I3); asked about allocating labor.

20 ASSISTANT

Confirmed the minimization objective. Clarified the optimizer, not the student, chooses the split between regular and overtime. Introduced R1,R2,R3 <= 2700; no finite overtime upper bound; P_t = R_t + O_t; and quarter-specific inventory equations.

21 USER

First attempted constraints, multiplying regular and overtime variables by $40/$60 and inventory by $15 in demand inequalities; requested feedback.

22 ASSISTANT

Explained that dollar coefficients belong in the objective, not physical capacity or inventory constraints. Derived the quarter 1 inventory balance I1 = 0.9[1000 + 0.8(R1+O1) - 3000].

23 USER

Asked for another explanation of inventory mechanics.

24 ASSISTANT

Walked through 3000 total units produced in Q1: 2400 usable new units + 1000 beginning inventory - 3000 demand = 400; 10% of 400 spoils; I1 = 360 carries into Q2. Explained Q2 begins with I1 rather than another 1000.

25 USER

Revised regular, overtime, and inventory constraints but initially omitted the previous quarter's ending inventory in Q2 and Q3.

26 ASSISTANT

Confirmed regular capacity and Q1 inventory. Explained Q2 must include I1, Q3 must include I2, and the production-defining equalities must be supplied.

27 USER

Submitted a complete formulation.

```
R1 <= 2700; R2 <= 2700; R3 <= 2700
P1 = R1 + O1; P2 = R2 + O2; P3 = R3 + O3
I1 = 0.9(1000 + 0.8P1 - 3000)
I2 = 0.9(I1 + 0.8P2 - 2000)
I3 = 0.9(I2 + 0.8P3 - 4000)
All R, O, I, P >= 0
```

28 ASSISTANT

Verified the formulation and noted that overtime only requires nonnegativity; corrected typing slips O30/P30 to O3/P3.

29 USER

Asked whether already-written equations could be moved into assignment sections headed Production Balance Constraints and Inventory Dynamics.

30 ASSISTANT

Confirmed the three P equalities and three I equalities were exactly the required content for those sections.

31 USER

Moved to R, asked whether make.lp(0,12) was correct for four types of variable in three quarters.

32 ASSISTANT

Confirmed 12 variables and fixed order: R1,R2,R3,O1,O2,O3,I1,I2,I3,P1,P2,P3.

33 USER

Asked how to replace zeros in a twelve-entry coefficient vector for regular-time capacity.

34 ASSISTANT

Explained one-hot coefficients for R1 <= 2700: 1 at position 1, zeros elsewhere; invited student to produce the R2 line.

35 USER

Submitted three regular capacity constraints and explicit nonnegativity constraints for overtime.

```
add.constraint(ProductMix.LP,c(1,0,0,0,0,0,0,0,0,0,0,0),"<=",2700)
add.constraint(ProductMix.LP,c(0,1,0,0,0,0,0,0,0,0,0,0),"<=",2700)
add.constraint(ProductMix.LP,c(0,0,1,0,0,0,0,0,0,0,0,0),"<=",2700)
add.constraint(ProductMix.LP,c(0,0,0,1,0,0,0,0,0,0,0,0),">=",0)
add.constraint(ProductMix.LP,c(0,0,0,0,1,0,0,0,0,0,0,0),">=",0)
add.constraint(ProductMix.LP,c(0,0,0,0,0,1,0,0,0,0,0,0),">=",0)
```

36 ASSISTANT

Confirmed and explained that production equalities use "=" and zero RHS after rearrangement, e.g., -R1-O1+P1=0.

37 USER

Submitted all three production-balance add.constraint lines.

```
add.constraint(ProductMix.LP,c(-1,0,0,-1,0,0,0,0,0,1,0,0),"=",0)
add.constraint(ProductMix.LP,c(0,-1,0,0,-1,0,0,0,0,0,1,0),"=",0)
add.constraint(ProductMix.LP,c(0,0,-1,0,0,-1,0,0,0,0,0,1),"=",0)
```

38 ASSISTANT

Verified all three lines and continued to inventory dynamics.

39 USER

Asked how to incorporate constants such as 0.8 into add.constraint coefficient vectors.

40 ASSISTANT

Expanded I1 = 0.9(1000+0.8P1-3000) into I1 - 0.72P1 = -1800; R vector has +1 at position 7 and -0.72 at position 10, RHS -1800.

41 USER

Attempted the Q2 inventory constraint but initially missed I1.

```
add.constraint(ProductMix.LP,c(0,0,0,0,0,0,0,1,0,0,-0.72,0),"=",-1800)
```

42 ASSISTANT

Explained Q2 must be I2 - 0.9I1 - 0.72P2 = -1800, and that -1800 is the correct constant.

43 USER

Corrected Q2.

```
add.constraint(ProductMix.LP,c(0,0,0,0,0,0,-0.9,1,0,0,-0.72,0),"=",-1800)
```

44 ASSISTANT

Confirmed Q2 coefficient placements and RHS. Asked for Q3.

45 USER

Attempted Q3 with the correct coefficient vector but RHS -1800.

```
add.constraint(ProductMix.LP,c(0,0,0,0,0,0,0,-0.9,1,0,0,-0.72),"=",-1800)
```

46 ASSISTANT

Corrected only Q3 RHS to -3600, since 0.9 x 4000 = 3600.

47 USER

Proposed objective coefficients, asking whether P terms should be zero.

```
set.objfn(ProductMix.LP,c(40,40,40,60,60,60,15,15,15,0,0,0))
```

48 ASSISTANT

Confirmed P costs are zero because production costs already enter through R and O. Reminded user to set lp.control(...,sense="min").

49 USER

Reported implausible solver output: objective -1e+30 and enormous values such as 1.175496e+214.

50 ASSISTANT

Identified that the LP may not have solved optimally; advised checking solve() status and model printout, and checking bounds and minimization direction.

51 USER

Pasted the full R script. It had duplicate regular/overtime constraint rows and lp.control(ProductMix.LP,sense="max") while costs were supposed to be minimized.

52 ASSISTANT

Pinpointed the main error: change "max" to "min". Flagged duplicate regular production/overtime rows as redundant. Recommended rerunning the entire script from make.lp(0,12).

53 USER

Reported corrected solver results.

```
get.objective(ProductMix.LP)
[1] 445360
get.variables(ProductMix.LP)
[1] 2500 2700 2700 0 0 2120 0 144 0 2500 2700 4820
```

54 ASSISTANT

Interpreted minimum total cost as $445,360. Q1: R 2500, O 0, I 0, P 2500; Q2: R 2700, O 0, I 144, P 2700; Q3: R 2700, O 2120, I 0, P 4820. Verified cost: $316,000 regular + $127,200 overtime + $2,160 inventory = $445,360.

55 USER

Summarized optimum correctly but accidentally labeled the 2120 overtime units as Q1O rather than Q3O.

56 ASSISTANT

Confirmed all numerical results and corrected the single label to Q3O: 2120.

57 USER

Requested: "THANK YOU. PLEASE RENDER A PDF OF OUR SESSION LOG".

Final verified solution

```
Minimum cost: $445,360
Quarter       1       2       3
Regular       2500    2700    2700
Overtime         0       0    2120
Ending Inv.      0     144       0
Total Prod.   2500    2700    4820
```

End of reconstructed session log.